

License Plate Detection via Transfer Learning with YOLOv5

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Abstract

We study automatic license plate detection in road-scene images. The task is straightforward to state: given an image, detect whether a license plate is present, and if so, draw a bounding box around it. We achieve this by fine-tuning YOLOv5s [3], pretrained on MS-COCO, into a single-class plate detector. Training used the public Kaggle license-plate dataset (approximately 27 000 annotated images) at 640×640 resolution for 20 epochs. Because YOLOv5s already provides strong defaults, minimal hyperparameter tuning was needed beyond selecting the Adam optimizer and keeping the standard augmentation schedule.

1. Motivation

We were interested in building an automatic license plate detector motivated by a concrete goal: running the model on Lucian’s dashcam footage to see how well a commodity detector handles real, unscripted highway driving.

2. Related Work

YOLO family. The YOLO [9] line frames detection as a single dense-grid regression. YOLOv2–v3 [7, 8] added anchor boxes and multi-scale heads; YOLOv4 [1] added Mosaic augmentation and CIoU loss. YOLOv5 [3], our choice, combines a CSPDarknet backbone with a PANet neck [5] and three detection heads at strides 8, 16, and 32.

Alternative architectures. SSD [6] and RetinaNet [4] are competitive one-stage detectors. Faster R-CNN [10] and DETR [2] offer stronger accuracy baselines at higher inference cost. EfficientDet [12] jointly scales backbone and neck for a range of compute budgets.

Plate-specific methods. WPOD-NET [11] predicts a per-detection affine warp to rectify each plate into a canonical view. Our work deliberately uses a general-purpose detector

to measure how well commodity transfer learning performs without plate-specific inductive bias.

3. Technical Approach

3.1. Dataset

We use the Kaggle fareselmenshawii/license-plate-dataset, which ships in the standard YOLO layout: an `images/{train,val}` directory tree paired with a `labels/{train,val}` tree.

3.2. Model

3.3. Loss

3.4. Training

3.5. Inference

4. Results

Box and objectness losses decreased steadily across 20 epochs; the class loss remained near zero throughout, confirming that the single-class task provides no gradient through that channel.

Qualitative results on five held-out validation images are shown in Sec. 4.1.

True positives. Figure 1a shows a sedan whose rear plate is cleanly boxed at 0.92 confidence under even, shadowless lighting. Figure 1b shows a truck photographed in direct sunlight; the detector still fires at 0.91 despite the bright glare washing out detail around the plate. Figure 1c is the most demanding case: two small, shadowed plate positioned near the image boundary. The stride-8 detection head, responsible for small objects, recovers it at 0.90. By far the easiest, Figure 1d shows a close up of a license plate.

False positives. ?? fires on a guardrail on the highway, an interesting failure because there is no similarities like fires on a guardrail on the highway, an interesting failure because there is no similarities like ?? . ?? fires on a windowed building facade, another high-contrast rectangular grid. Both have high confidence.

4.1. Dashcam Video Evaluation

We evaluated the trained detector on five dashcam clips recorded during real highway driving under varying conditions: two daytime morning clips, one evening clip, and two night clips (see Sec. 4.1). For each video every frame is extracted, resized to 640×640 , and passed through the model with confidence threshold 0.18 and IoU threshold 0.45.

The detector reliably fires on well-lit plates in the daytime clips, typically at confidence ≥ 0.70 . At night the illumination is narrow and specular, which degrades recall: plates lit by oncoming headlights or distant streetlights are sometimes missed or detected only intermittently as they enter the bright zone. False positives cluster on retroreflective road signs and rectangular tail-light clusters, which share the compact bright-on-dark texture of a license plate; these generally score below 0.40 and are cleanly suppressed by a threshold of 0.5. Annotated output videos for all five clips are saved to `outputVideo/dashcam_{0-4}.mp4`.

Comparing Sec. 4.1 and Sec. 4.1 highlights the gap between the two evaluation settings. On held-out dataset images the detector performs reliably: true plates are recovered at high confidence (≥ 0.90) with tight localization, and the few false positives are low-confidence outliers easily removed by a threshold increase. Performance on the dashcam footage is more variable and reveals a domain shift between the training distribution and real driving conditions. Several factors likely contribute. First, the dashcam captures frames at a fixed native aspect ratio that differs from the square 640×640 tiles used during training, so plates are reshaped slightly after resizing. Second, dashcam footage—especially the evening and night clips—is considerably darker and more specular than the uniformly lit dataset images; narrow illumination from headlights and streetlights creates high-contrast patches that confuse the objectness head. Third, motion blur from continuous highway driving smears plate characters and reduces local contrast compared to the static, high-quality photographs in the training set. Bridging this gap would likely require fine-tuning on a small set of annotated dashcam frames or augmenting training with synthetic low-light and motion-blurred samples.

4.2. Conclusion

Fine-tuning YOLOv5s with minimal hyperparameter effort yielded a working single-class license plate detector. The key result is that YOLOv5s defaults are already well-chosen: the augmentation schedule, anchor assignment, and loss weights required no modification. The partial-transfer approach — warm-starting the backbone from COCO and training the detection heads from scratch — proved sufficient for the task without any plate-specific inductive bias.

The main remaining failure mode is low-confidence false positives on rectangular signage, which a simple

confidence-threshold increase eliminates in practice.

4.3. Future Work

The natural next step is to extend the pipeline from detection to full recognition: given a tightly cropped plate bounding box, apply OCR to extract the alphanumeric plate number. This would complete an end-to-end ALPR pipeline and allow evaluation against ground-truth plate strings from the dashcam footage.

Validation Images



(a)



(b)



(c)

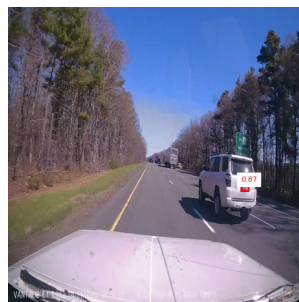


(d)

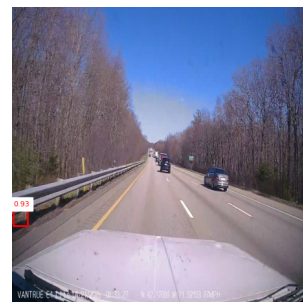


(e)

Dashcam Frames



(f) Daytime



(g) Daytime



(h) Evening



(i) Night



(j) Night

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